



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CENTRE
NUMBER

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INDEX
NUMBER

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CLASS

H2 BIOLOGY

9648/03

Applications Paper and Planning Question

26 September 2013

Paper 3

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your centre number, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten ruled paper for Question 5 separately.

Hand in the Question Paper, answers to Question 5 separately.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/ 15
2	/ 12
3	/ 13
4 (SPA)	/ 12
5	/ 20
Total	

This Question Paper consists of **17** printed pages.

Answer **all** questions.

1. DNA fingerprinting is a method of identification based on an individual's DNA. In traditional DNA fingerprinting, DNA is first isolated from cells, and then amplified by Polymerase Chain Reaction (PCR). **Fig. 1.1** illustrates the process of PCR.

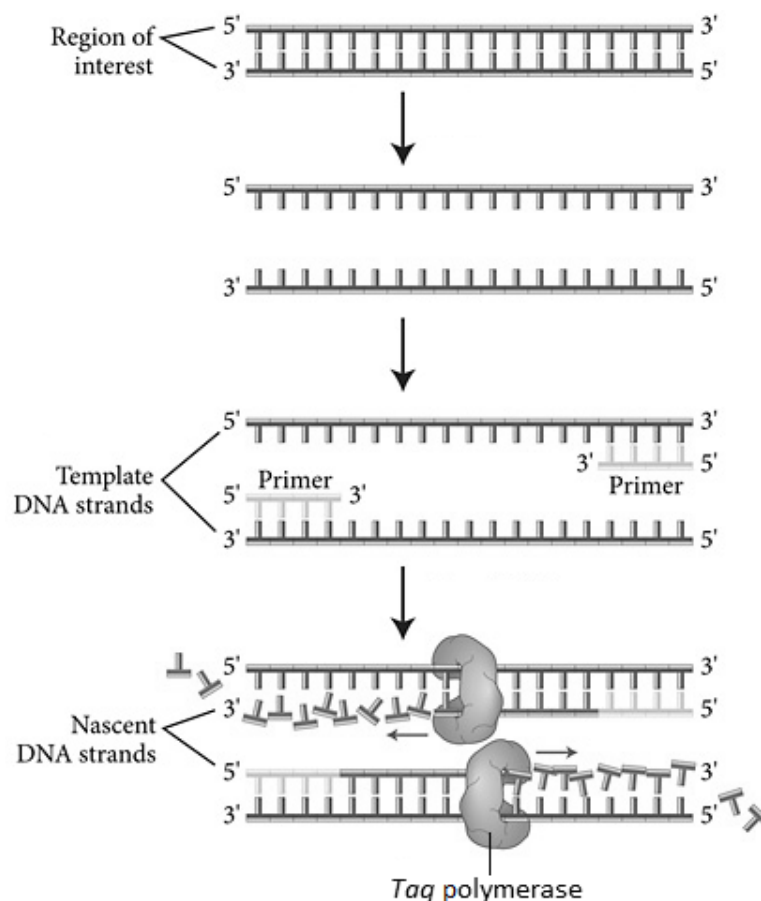


Fig. 1.1

- (a) (i) *Taq* polymerase is derived from *Thermus aquaticus*, a species of bacteria that thrives in hot springs. Explain why *Taq* polymerase is able to increase the amount of DNA from most species of organisms. [2]

1. (a) (ii) Explain how the specific nucleotide sequence of the primer is important for its role in PCR. [2]

- (b) Distinguish between PCR and DNA replication in human skin cells. [2]

1. DNA amplified using PCR is then cut by a specific restriction enzyme and separated using gel electrophoresis. The DNA is then transferred to a nitrocellulose membrane, before the addition of radioactive DNA probes for nucleic acid hybridisation. **Fig. 1.2** shows the nucleic acid hybridisation process.

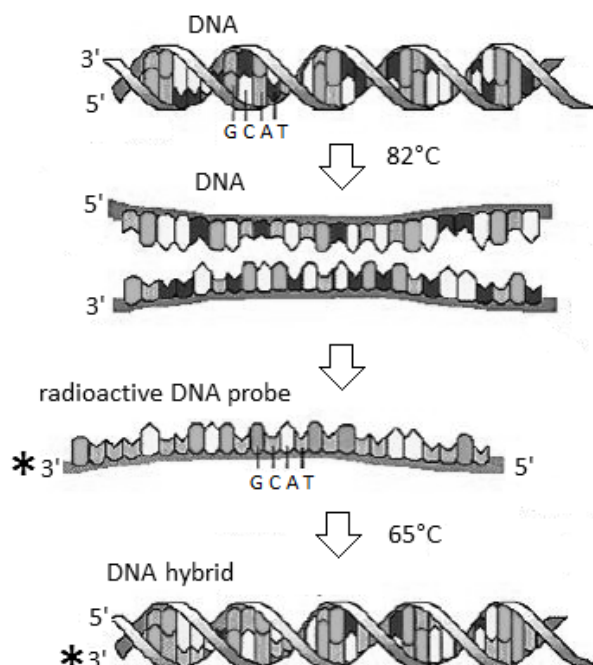


Fig. 1.2

1. (c) With reference to **Fig. 1.2**, explain the significance of the temperatures used during nucleic acid hybridisation. [3]

1. (d) Suggest why several probes have to be used in order for DNA fingerprinting to be effective. [2]

DNA fingerprinting is often used in paternity testing. This involves the investigation of various Variable Number of Tandem Repeat (VNTR) loci in the individual's genome. The table below shows the genotype of five individuals at each of the five loci examined.

	Locus on chromosome				
	D19S433	D3S1358	SE33	D8S1179	D21S11
Mother	16,16	15,16	20,22	13,14	28,30
Child	16,18	16,16	22,24	13,14	28,29
Male 1	16,18	16,16	22,23	14,15	28,31
Male 2	17,18	15,16	24,25	11,12	28,30
Male 3	15,18	16,16	19,24	12,13	29,31

- (e) Define *Variable Number of Tandem Repeat*. [2]

- (f) (i) State the male(s) who may possibly be the father of the child. [1]

1. (f) (ii) Explain why the test above cannot conclude with absolute certainty that a given male is the father of the child. [1]

[Total: 15]

2. In humans, the gene *RPE65* encodes a protein responsible for regenerating visual pigment in rod and cone cells after they have been exposed to light. A recessive allele of this gene causes impaired vision from birth, progressing to complete blindness in early adulthood. This condition is called LCA.

In 2008, trials were carried out into the possibility and safety of treating LCA using gene therapy.

- (a) Suggest and explain why LCA is suitable for treatment using gene therapy. [3]

Six adults with this condition were used in the study. Genetically modified adenoviruses (a type of virus that can cause respiratory infections) were used as vectors. The vectors were injected beneath the retina of one eye of each of the participants.

- (b) Suggest two ways in which the genome of the adenoviruses used as vectors would differ from that of normal adenoviruses. [2]

2. Improvements were found in the vision of all the participants, but the small number in the trials made most of these improvements not statistically significant.

(c) Suggest why these trials were designed to include such a small number of participants.

[2]

Scientists have developed stem cell therapy for retinal disease in mouse. The adult stem cells from donors form functional photoreceptors after transplantation in the adult diseased mouse retina. The immune system of the mouse was suppressed to prevent them from rejecting the foreign stem cells.

(d) Adult stem cells such as these are described as multipotent.

Explain what is meant by the term *multipotent*.

[1]

(e) Suggest how it is possible that adult stem cells could differentiate into the range of cell types needed for repairing retina.

[2]

2. (f) Suggest the advantages of using stem cells from donors as compared to the use of adenoviruses in (b). [2]

[Total: 12]

3. Rice is the staple diet in many parts of the world. Rice plants produce the precursor of vitamin A, β -carotene, in their green tissues, but not in the edible endosperms of their seeds. Adequate concentrations of vitamin A give protection from night blindness. Higher concentrations act as an antioxidant that may give some protection from cancer and heart disease.

Golden rice, which contains β -carotene, was developed in Switzerland by genetically modifying rice using genes from a daffodil (a flowering plant) and a bacterium. The rice produces seeds containing β -carotene.

Fig. 3.1 shows the DNA sequence used.



Key

pro - start site for polymerase enzymes

ter - end signal for polymerase enzymes

psy - gene from daffodil (a flowering plant)

crt 1 - gene from bacterium

Hyg resist - antibiotic resistance gene from a bacterium

Fig. 3.1

The following steps were taken to produce golden rice:

- Step 1:** a length of DNA (shown in **Fig. 3.1**) was made including a rice endosperm-specific promoter, *psy* gene, *crt 1* gene and *Hyg resist* gene,
- Step 2:** copies of this length of DNA were inserted into plasmids from the bacterium *Agrobacterium tumefaciens*,
- Step 3:** *A. tumefaciens* containing the plasmids were mixed with rice embryos and cultured in a medium containing the antibiotic *Hyg*,
- Step 4:** the embryos growing successfully on *Hyg* were grown into plantlets and then plants.

- (a) Describe the role of the rice endosperm-specific promoter that was added to *psy* and *crt 1* in **Step 1**.

[2]

3. (b) Explain how a length of DNA can be inserted into a plasmid in **Step 2**. [3]

- (c) Explain the role of the *Hyg* resistance gene in this procedure. [1]

Field trials of genetically engineered crops, such as golden rice, hope to identify any risks to the environment or to human health of growing and eating the crops.

- (d) Suggest:

- (i) one possible risk to the environment of growing a genetically engineered crop; [1]

- (ii) one possible risk to human health of eating a genetically engineered crop. [1]

3. A group of adult volunteers were given a 500 g of labelled golden rice dose for three days together with 10 g butter. Each gram of golden rice contains 1.6 ug of β -carotene. Serum samples from the volunteers were analysed for golden rice β -carotene. The result of the analysis was shown in **Fig. 3.2**.

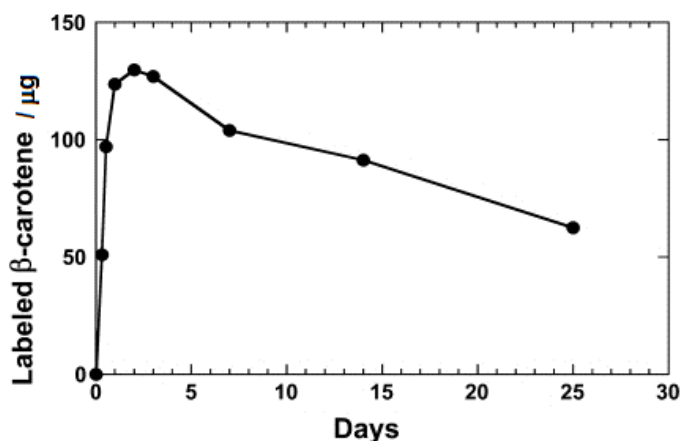


Fig. 3.2

- (e) (i) With reference to **Fig. 3.2**, explain the change in labelled β -carotene. [2]

- (ii) Suggest why the volunteers need to consume 10 g of butter together with the labelled golden rice. [1]

- (iii) Suggest why children are not included in the study. [1]

3. The daily recommended intake of β -carotene is 2 mg.

- (f) Suggest if the daily intake of golden rice by children will overcome the problem of vitamin A deficiency. Explain your answer. [1]

[Total: 13]

4. Planning Question

You are required to plan, but **not** carry out, an investigation into the effect of the independent variable, copper sulphate concentration, on amylase.

Amylase catalyses the hydrolysis of starch to glucose.

Copper sulfate can inhibit amylase.

The extent of inhibition depends on the concentration of the copper sulfate solution. A student investigated the inhibition of amylase at concentrations of copper sulfate solution greater than 3.0% and found that the enzyme was completely inhibited.

The student suggested the hypothesis:

Concentrations of copper sulfate solution below 3.0% will continue to inhibit the enzyme.

You are required to investigate this hypothesis by carrying out a serial dilution of copper sulfate solution which reduces the concentration by ten-fold between each successive dilution.

Using this information and your own knowledge, design an experiment to test the student's hypothesis.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- 1% amylase solution,
- 1% starch suspension,
- copper sulfate (powder),
- iodine in potassium iodide solution,
- any normal laboratory glassware, e.g. test-tubes, beakers, measuring cylinders, glass rod, etc.,
- syringes,
- timer e.g. stopwatch,
- colourimeter,
- cuvette.

[Total: 12]

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.

[illegible]

